



VISION

CODESLAYER 2K26

DevSphereIndia



Project Title : CareLink — Healthcare Continuity and Referral Intelligence Platform

Team Name : VISION

Team Members : Kartikey Sharma - Research & Presentation

Shivam Rai - Full Stack & AI Integration

Sakshi Patwa - Frontend & UI/UX Design

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Tracks Chosen: Healthcare + AI/ML

PROBLEM STATEMENT & TARGET AUDIENCE

PROBLEM WE'RE SOLVING

India's healthcare system is losing patients to broken handoffs between providers.

- **Referral gap:** 25,000+ PHCs create referrals daily, but nobody tracks if the patient arrived.
- **Diagnostic gap:** critical abnormal reports can sit unreviewed for 8+ days; no automated follow-up fires.
- **Medication gap:** multi-provider conflicts can go undetected across siloed systems.
- **Access gap:** patients travel 60km without knowing whether the required specialist or equipment is available.
- **Visibility gap:** respiratory spikes across 12 OPDs remain invisible when every hospital sees only its own data.

WHO 2025: digital referral dropout and follow-up remain unsolved after digitization.

Capacity pressure: India's doctor-to-population ratio is 1:834; staff cannot manually track every handoff.

TARGET USERS

- **PHC & district doctors:** creating and managing referrals (B2B).
- **ASHA workers:** coordinating patient arrival and follow-up.
- **District health officers:** monitoring population signals and throughput.
- **Pharmacists:** reconciling multi-provider medication histories.
- **Patients (B2C):** viewing one unified health journey.

Scale of impact

1.6 lakh+ public health facilities · 25,000+ referral-creating PHCs
· MP pilot target: 1,273 PHCs.



THE COORDINATION LAYER ABOVE EXISTING SYSTEMS

CareLink is not a health record or telemedicine app. It tracks what happens to the patient between handoffs, detects broken workflow states, and recommends the next human-approved action.



Referral Risk Scorer

Explainable 0–100 score using distance, missed visits, transport and facility load.

Patient Timeline + RAG

OCR documents into a single journey; answers cite the source record.

Diagnostic Follow-up

Ordered → reviewed → informed → action taken; alerts on missed windows.

Medication Reconciliation

Normalizes multi-provider drugs; OpenFDA checks conflicts; pharmacist decides.

WHY IT'S DIFFERENT

- ✓ **ABDM** stores records → CareLink detects when the journey breaks.
- ✓ **eSanjeevani** enables teleconsults → CareLink matches a capable facility now.
- ✓ **Referral systems** create referrals → CareLink tracks every stage after creation.
- ✓ **Hospital EHRs** store results → CareLink escalates unreviewed critical results.
- ✓ **Drug apps** check one prescription → CareLink reconciles across the full journey.

CareBot agents	What they watch	Human outcome
Referral + Risk	Dropout & facility fit	Call / book / escalate
Diagnostic	Result review delay	Clinician action
Medication	Drug discrepancy	Pharmacist review

Two-point novelty: a care-gap state machine and an explainable dropout score shown at referral creation. **AI flags. Humans decide.**



TECH STACK

Frontend

React 18 + Vite · TailwindCSS · Recharts ·
Framer Motion

Backend

Node.js · Express.js · JWT + bcrypt · node-cron ·
validation

AI & APIs

OpenAI GPT-4o · LangChain RAG · OpenFDA ·
Twilio SMS

Data & Infra

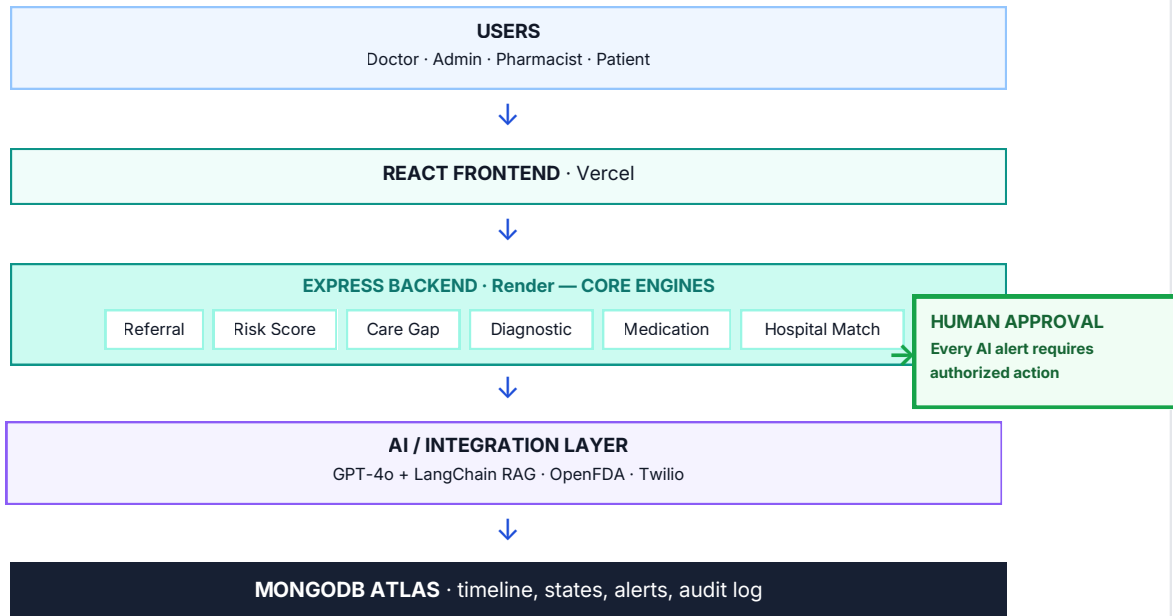
MongoDB Atlas · Vercel · Render · GitHub CI/CD

Prototype data

Synthetic de-identified patients, referrals and
MP facility feeds; production path via HMIS /
ABDM.

Architecture principle: deterministic rules detect
gaps; AI explains and summarizes; authorized
humans approve consequential action.

ARCHITECTURE



Referral data flow: create → score → match facility → monitor every 4h → alert → advance stage → timeline + analytics



FEASIBILITY

- ✓ **Free-tier ready:** MongoDB Atlas, OpenFDA, Vercel, Render and GitHub support a low-cost prototype.
- ✓ **Modular build:** 7 features demo independently; incomplete modules do not break the core pitch.
- ✓ **Fast demo setup:** seed script creates 3 hospitals, 5 patients, 8 referrals, 10 results and 3 conflicts in 30 seconds.
- ✓ **Rule-based risk:** transparent scoring works on day one — no ML training data required.
- ✓ **Cloud + offline:** local MongoDB and Vite fallback keep the demo running without venue connectivity.

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team roles

36h

hackathon-ready

₹200

demo API budget

Resources: frontend, backend, AI integration and research/presentation ownership; no hardware dependency.

SHOWSTOPPERS → MITIGATIONS

OpenFDA rate limits / downtime

Cache responses in MongoDB; preload common drug pairs; label fallback clearly.

OpenAI latency / cost

Cache classifications and use non-streaming CareBot mode; budget under ₹200.

"Where is real-time hospital data?"

Demo uses clearly labelled synthetic MP feeds; production path is HMIS / ABDM integration.

"Is AI replacing doctors?"

CareLink tracks workflow states, not diagnoses. Human approval is mandatory.

Network failure at venue

Run local MongoDB + local backend; pre-build and test offline mode.

Trust position: AI flags missed workflow steps, explains evidence and suggests action — **humans decide.**

USP(UNIQUE SELLING PRICE) & BUSINESS MODEL



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USP — WHY CARELINK WINS

- ◆ **Gap, not transaction:** existing platforms store records or complete one task; CareLink owns the space between handoffs.
- ◆ **Explainable by design:** every risk point, drug source and diagnostic finding is visible — never “AI said so.”
- ◆ **Works with infrastructure:** integrates with ABDM, state referral systems and HMIS instead of replacing them.
- ◆ **Proactive intelligence:** predicts where the journey will fail and alerts before a patient disappears.

“Not a health record. Not a telemedicine app. The coordination layer around the patient.”

Long-term: the API layer connecting ABDM records, eSanjeevani and state referrals into one continuous journey.

BUSINESS MODEL & SCALE

Phase 1 Months 1–6

Free pilot with 2–3 government district hospitals in Madhya Pradesh.

Impact data

Phase 2 Months 6–18

Government SaaS at ₹15–25 / bed / month for district hospitals.

₹9.4–15.6L
MP potential

Phase 3 Year 2+

NHM national integration; private-chain integration APIs.

₹1.5Cr / month
PHC potential

Revenue streams: government subscriptions · private integration APIs · district analytics · training & implementation.

Why it pays: preventing even 10 readmissions per hospital per month can offset the cost of deployment.

REFERENCES

WHO 2025 Karnataka referral case study · ABDM documentation · OpenFDA API · NHM Health Facility Registry · PubMed studies on medication reconciliation · WHO Patient Safety Goals 2025.